

1 Task 1.6:

полный вывод jstack, ключевые фрагменты и короткое объяснение, почему дамп подтверждает дедлок.

2 one_thread

вывод jstack

```
PS C:\Users\dan> jps
49268 one_thread
50088 Jps
50248 Main
59848 Launcher
PS C:\Users\dan> jstack 49268
2026-02-23 04:34:17
Full thread dump OpenJDK 64-Bit Server VM (11.0.30+7-LTS mixed mode, sharing):

Threads class SMR info:
_java_thread_list=0x000001b89b29ff80, length=11, elements={
0x000001b8f6417800, 0x000001b89a763800, 0x000001b89a766800, 0x000001b89a783000,
0x000001b89a784000, 0x000001b89a785800, 0x000001b89a788800, 0x000001b89a78b800,
0x000001b89a78c800, 0x000001b89b030800, 0x000001b89b36f800
}

"main" #1 prio=5 os_prio=0 cpu=0.00ms elapsed=39.72s tid=0x000001b8f6417800 nid=0xdee4 in
  Object.wait() [0x000000acbabff000]
    java.lang.Thread.State: WAITING (on object monitor)
      at java.lang.Object.wait(java.base@11.0.30/Native Method)
        - waiting on <0x000000071a107ed8> (a java.lang.Thread)
      at java.lang.Thread.join(java.base@11.0.30/Thread.java:1300)
        - waiting to re-lock in wait() <0x000000071a107ed8> (a java.lang.Thread)
      at java.lang.Thread.join(java.base@11.0.30/Thread.java:1375)
      at one_thread.main(one_thread.java:3)

"Reference Handler" #2 daemon prio=10 os_prio=2 cpu=0.00ms elapsed=39.70s tid=0
  x000001b89a763800 nid=0xc61c waiting on condition [0x000000acbb2ff000]
    java.lang.Thread.State: RUNNABLE
      at java.lang.ref.Reference.waitForReferencePendingList(java.base@11.0.30/Native
        Method)
      at java.lang.ref.Reference.processPendingReferences(java.base@11.0.30/Reference.java
        :241)
      at java.lang.ref.Reference$ReferenceHandler.run(java.base@11.0.30/Reference.java
        :213)

"Finalizer" #3 daemon prio=8 os_prio=1 cpu=0.00ms elapsed=39.70s tid=0x000001b89a766800 nid
  =0xca0c in Object.wait() [0x000000acbb3ff000]
    java.lang.Thread.State: WAITING (on object monitor)
      at java.lang.Object.wait(java.base@11.0.30/Native Method)
        - waiting on <0x000000071a10a2e0> (a java.lang.ref.ReferenceQueue$Lock)
      at java.lang.ref.ReferenceQueue.remove(java.base@11.0.30/ReferenceQueue.java:155)
        - waiting to re-lock in wait() <0x000000071a10a2e0> (a java.lang.ref.
        ReferenceQueue$Lock)
      at java.lang.ref.ReferenceQueue.remove(java.base@11.0.30/ReferenceQueue.java:176)
      at java.lang.ref.Finalizer$FinalizerThread.run(java.base@11.0.30/Finalizer.java:170)

"Signal Dispatcher" #4 daemon prio=9 os_prio=2 cpu=0.00ms elapsed=39.69s tid=0
  x000001b89a783000 nid=0xdd28 runnable [0x0000000000000000]
```

```

java.lang.Thread.State: RUNNABLE

"Attach Listener" #5 daemon prio=5 os_prio=2 cpu=0.00ms elapsed=39.69s tid=0
  x000001b89a784000 nid=0xa564 waiting on condition [0x0000000000000000]
java.lang.Thread.State: RUNNABLE

"Service Thread" #6 daemon prio=9 os_prio=0 cpu=0.00ms elapsed=39.69s tid=0x000001b89a785800
  nid=0xcdcc runnable [0x0000000000000000]
java.lang.Thread.State: RUNNABLE

"C2 CompilerThread0" #7 daemon prio=9 os_prio=2 cpu=15.62ms elapsed=39.69s tid=0
  x000001b89a788800 nid=0xa68c waiting on condition [0x0000000000000000]
java.lang.Thread.State: RUNNABLE
No compile task

"C1 CompilerThread0" #15 daemon prio=9 os_prio=2 cpu=0.00ms elapsed=39.69s tid=0
  x000001b89a78b800 nid=0xcb20 waiting on condition [0x0000000000000000]
java.lang.Thread.State: RUNNABLE
No compile task

"Sweeper thread" #19 daemon prio=9 os_prio=2 cpu=0.00ms elapsed=39.69s tid=0
  x000001b89a78c800 nid=0xa87c runnable [0x0000000000000000]
java.lang.Thread.State: RUNNABLE

"Common-Cleaner" #20 daemon prio=8 os_prio=1 cpu=0.00ms elapsed=39.66s tid=0
  x000001b89b030800 nid=0xadfc in Object.wait() [0x000000acbbaff000]
java.lang.Thread.State: TIMED_WAITING (on object monitor)
  at java.lang.Object.wait(java.base@11.0.30/Native Method)
  - waiting on <0x000000071a008b80> (a java.lang.ref.ReferenceQueue$Lock)
  at java.lang.ref.ReferenceQueue.remove(java.base@11.0.30/ReferenceQueue.java:155)
  - waiting to re-lock in wait() <0x000000071a008b80> (a java.lang.ref.
ReferenceQueue$Lock)
  at jdk.internal.ref.CleanerImpl.run(java.base@11.0.30/CleanerImpl.java:148)
  at java.lang.Thread.run(java.base@11.0.30/Thread.java:829)
  at jdk.internal.misc.InnocuousThread.run(java.base@11.0.30/InnocuousThread.java:161)

"Monitor Ctrl-Break" #21 daemon prio=5 os_prio=0 cpu=0.00ms elapsed=39.61s tid=0
  x000001b89b36f800 nid=0x2774 runnable [0x000000acbbefe000]
java.lang.Thread.State: RUNNABLE
  at java.net.SocketInputStream.socketRead0(java.base@11.0.30/Native Method)
  at java.net.SocketInputStream.socketRead(java.base@11.0.30/SocketInputStream.java
:115)
  at java.net.SocketInputStream.read(java.base@11.0.30/SocketInputStream.java:168)
  at java.net.SocketInputStream.read(java.base@11.0.30/SocketInputStream.java:140)
  at sun.nio.cs.StreamDecoder.readBytes(java.base@11.0.30/StreamDecoder.java:295)
  at sun.nio.cs.StreamDecoder.implRead(java.base@11.0.30/StreamDecoder.java:337)
  at sun.nio.cs.StreamDecoder.read(java.base@11.0.30/StreamDecoder.java:179)
  - locked <0x0000000719f05290> (a java.io.InputStreamReader)
  at java.io.InputStreamReader.read(java.base@11.0.30/InputStreamReader.java:181)
  at java.io.BufferedReader.fill(java.base@11.0.30/BufferedReader.java:161)
  at java.io.BufferedReader.readLine(java.base@11.0.30/BufferedReader.java:326)
  - locked <0x0000000719f05290> (a java.io.InputStreamReader)
  at java.io.BufferedReader.readLine(java.base@11.0.30/BufferedReader.java:392)
  at com.intellij.rt.execution.application.AppMainV2$1.run(AppMainV2.java:31)

"VM Thread" os_prio=2 cpu=0.00ms elapsed=39.70s tid=0x000001b89a761800 nid=0x8c30 runnable

"GC Thread#0" os_prio=2 cpu=0.00ms elapsed=39.72s tid=0x000001b8f6433000 nid=0xcce4 runnable

```

```
"G1 Main Marker" os_prio=2 cpu=0.00ms elapsed=39.72s tid=0x000001b8f648c800 nid=0xf1f0
  runnable
"G1 Conc#0" os_prio=2 cpu=0.00ms elapsed=39.72s tid=0x000001b8f648e000 nid=0x4a80 runnable
"G1 Refine#0" os_prio=2 cpu=0.00ms elapsed=39.71s tid=0x000001b89a676000 nid=0xe3b4 runnable
"G1 Young RemSet Sampling" os_prio=2 cpu=0.00ms elapsed=39.71s tid=0x000001b89a677000 nid=0
  xbe48 runnable
"VM Periodic Task Thread" os_prio=2 cpu=0.00ms elapsed=39.61s tid=0x000001b89b370800 nid=0
  xa7b8 waiting on condition

JNI global refs: 24, weak refs: 0
```

```
"main"... java.lang.Thread.State: WAITING (on object monitor)
at java.lang.Thread.join(...)
at one_thread.main(one_thread.java:3)
```

почему это подтверждает дедлок

поток `main` находится в состоянии `WAITING` и блокируется внутри `Thread.join()` (`Thread.join(...)` и место вызова `one_thread.java:3`). вызывается `Thread.currentThread().join()`, таким образом поток дожидается завершения самого себя.

3 two_threads

Вывод jstack

```
PS C:\Users\dan> jps
15472 Jps
50248 Main
52268 Launcher
6604 two_threads1_6
PS C:\Users\dan> jstack 6603
6603: no such process
PS C:\Users\dan> jstack 6604
2026-02-23 04:37:59
Full thread dump OpenJDK 64-Bit Server VM (11.0.30+7-LTS mixed mode, sharing):

Threads class SMR info:
_java_thread_list=0x0000014b23f98df0, length=13, elements={
0x0000014b7f036000, 0x0000014b233e4000, 0x0000014b233e5800, 0x0000014b23402000,
0x0000014b23403000, 0x0000014b23404000, 0x0000014b23409800, 0x0000014b2340b000,
0x0000014b23425800, 0x0000014b23dd7000, 0x0000014b24027800, 0x0000014b24029000,
0x0000014b23f0c000
}

"main" #1 prio=5 os_prio=0 cpu=31.25ms elapsed=69.37s tid=0x0000014b7f036000 nid=0xcad0 in
  Object.wait() [0x0000003ce5aff000]
  java.lang.Thread.State: WAITING (on object monitor)
    at java.lang.Object.wait(java.base@11.0.30/Native Method)
      - waiting on <0x0000000719f62150> (a java.lang.Thread)
    at java.lang.Thread.join(java.base@11.0.30/Thread.java:1300)
      - waiting to re-lock in wait() <0x0000000719f62150> (a java.lang.Thread)
    at java.lang.Thread.join(java.base@11.0.30/Thread.java:1375)
    at two_threads1_6.main(two_threads1_6.java:23)

"Reference Handler" #2 daemon prio=10 os_prio=2 cpu=0.00ms elapsed=69.35s tid=0
  x0000014b233e4000 nid=0xf29c waiting on condition [0x0000003ce61ff000]
  java.lang.Thread.State: RUNNABLE
    at java.lang.ref.Reference.waitForReferencePendingList(java.base@11.0.30/Native
  Method)
    at java.lang.ref.Reference.processPendingReferences(java.base@11.0.30/Reference.java
  :241)
    at java.lang.ref.Reference$ReferenceHandler.run(java.base@11.0.30/Reference.java
  :213)

"Finalizer" #3 daemon prio=8 os_prio=1 cpu=0.00ms elapsed=69.34s tid=0x0000014b233e5800 nid
  =0x9e4c in Object.wait() [0x0000003ce62ff000]
  java.lang.Thread.State: WAITING (on object monitor)
    at java.lang.Object.wait(java.base@11.0.30/Native Method)
      - waiting on <0x000000071a10a2e0> (a java.lang.ref.ReferenceQueue$Lock)
    at java.lang.ref.ReferenceQueue.remove(java.base@11.0.30/ReferenceQueue.java:155)
      - waiting to re-lock in wait() <0x000000071a10a2e0> (a java.lang.ref.
  ReferenceQueue$Lock)
    at java.lang.ref.ReferenceQueue.remove(java.base@11.0.30/ReferenceQueue.java:176)
    at java.lang.ref.Finalizer$FinalizerThread.run(java.base@11.0.30/Finalizer.java:170)

"Signal Dispatcher" #4 daemon prio=9 os_prio=2 cpu=0.00ms elapsed=69.33s tid=0
  x0000014b23402000 nid=0x6704 runnable [0x0000000000000000]
  java.lang.Thread.State: RUNNABLE

"Attach Listener" #5 daemon prio=5 os_prio=2 cpu=0.00ms elapsed=69.33s tid=0
  x0000014b23403000 nid=0xee30 waiting on condition [0x0000000000000000]
```

```

java.lang.Thread.State: RUNNABLE

"Service Thread" #6 daemon prio=9 os_prio=0 cpu=0.00ms elapsed=69.33s tid=0x0000014b23404000
  nid=0xe0e0 runnable [0x0000000000000000]
java.lang.Thread.State: RUNNABLE

"C2 CompilerThread0" #7 daemon prio=9 os_prio=2 cpu=46.88ms elapsed=69.33s tid=0
  x0000014b23409800 nid=0xddb4 waiting on condition [0x0000000000000000]
java.lang.Thread.State: RUNNABLE
No compile task

"C1 CompilerThread0" #15 daemon prio=9 os_prio=2 cpu=0.00ms elapsed=69.33s tid=0
  x0000014b2340b000 nid=0x45a4 waiting on condition [0x0000000000000000]
java.lang.Thread.State: RUNNABLE
No compile task

"Sweeper thread" #19 daemon prio=9 os_prio=2 cpu=0.00ms elapsed=69.33s tid=0
  x0000014b23425800 nid=0xf2e0 runnable [0x0000000000000000]
java.lang.Thread.State: RUNNABLE

"Common-Cleaner" #20 daemon prio=8 os_prio=1 cpu=0.00ms elapsed=69.29s tid=0
  x0000014b23dd7000 nid=0xb5b4 in Object.wait() [0x0000003ce69ff000]
java.lang.Thread.State: TIMED_WAITING (on object monitor)
  at java.lang.Object.wait(java.base@11.0.30/Native Method)
  - waiting on <0x000000071a008b80> (a java.lang.ref.ReferenceQueue$Lock)
  at java.lang.ref.ReferenceQueue.remove(java.base@11.0.30/ReferenceQueue.java:155)
  - waiting to re-lock in wait() <0x000000071a008b80> (a java.lang.ref.
ReferenceQueue$Lock)
  at jdk.internal.ref.CleanerImpl.run(java.base@11.0.30/CleanerImpl.java:148)
  at java.lang.Thread.run(java.base@11.0.30/Thread.java:829)
  at jdk.internal.misc.InnocuousThread.run(java.base@11.0.30/InnocuousThread.java:161)

"Monitor Ctrl-Break" #21 daemon prio=5 os_prio=0 cpu=0.00ms elapsed=69.22s tid=0
  x0000014b24027800 nid=0x46f8 runnable [0x0000003ce6dfe000]
java.lang.Thread.State: RUNNABLE
  at java.net.SocketInputStream.socketRead0(java.base@11.0.30/Native Method)
  at java.net.SocketInputStream.socketRead(java.base@11.0.30/SocketInputStream.java
:115)
  at java.net.SocketInputStream.read(java.base@11.0.30/SocketInputStream.java:168)
  at java.net.SocketInputStream.read(java.base@11.0.30/SocketInputStream.java:140)
  at sun.nio.cs.StreamDecoder.readBytes(java.base@11.0.30/StreamDecoder.java:295)
  at sun.nio.cs.StreamDecoder.implRead(java.base@11.0.30/StreamDecoder.java:337)
  at sun.nio.cs.StreamDecoder.read(java.base@11.0.30/StreamDecoder.java:179)
  - locked <0x0000000719f88ba0> (a java.io.InputStreamReader)
  at java.io.InputStreamReader.read(java.base@11.0.30/InputStreamReader.java:181)
  at java.io.BufferedReader.fill(java.base@11.0.30/BufferedReader.java:161)
  at java.io.BufferedReader.readLine(java.base@11.0.30/BufferedReader.java:326)
  - locked <0x0000000719f88ba0> (a java.io.InputStreamReader)
  at java.io.BufferedReader.readLine(java.base@11.0.30/BufferedReader.java:392)
  at com.intellij.rt.execution.application.AppMainV2$1.run(AppMainV2.java:31)

"Thread-0" #22 prio=5 os_prio=0 cpu=0.00ms elapsed=69.21s tid=0x0000014b24029000 nid=0xb6cc
  in Object.wait() [0x0000003ce6fff000]
java.lang.Thread.State: WAITING (on object monitor)
  at java.lang.Object.wait(java.base@11.0.30/Native Method)
  - waiting on <0x0000000719f65220> (a java.lang.Thread)
  at java.lang.Thread.join(java.base@11.0.30/Thread.java:1300)
  - waiting to re-lock in wait() <0x0000000719f65220> (a java.lang.Thread)
  at java.lang.Thread.join(java.base@11.0.30/Thread.java:1375)
  at two_threads1_6.lambda$main$0(two_threads1_6.java:10)

```

```

        at two_threads1_6$$Lambda$14/0x0000000840064840.run(Unknown Source)
        at java.lang.Thread.run(java.base@11.0.30/Thread.java:829)

"Thread-1" #23 prio=5 os_prio=0 cpu=0.00ms elapsed=69.21s tid=0x0000014b23f0c000 nid=0xa4a4
  in Object.wait() [0x0000003ce70ff000]
  java.lang.Thread.State: WAITING (on object monitor)
    at java.lang.Object.wait(java.base@11.0.30/Native Method)
      - waiting on <0x0000000719f62150> (a java.lang.Thread)
    at java.lang.Thread.join(java.base@11.0.30/Thread.java:1300)
      - waiting to re-lock in wait() <0x0000000719f62150> (a java.lang.Thread)
    at java.lang.Thread.join(java.base@11.0.30/Thread.java:1375)
    at two_threads1_6.lambda$main$1(two_threads1_6.java:16)
    at two_threads1_6$$Lambda$15/0x0000000840064c40.run(Unknown Source)
    at java.lang.Thread.run(java.base@11.0.30/Thread.java:829)

"VM Thread" os_prio=2 cpu=0.00ms elapsed=69.35s tid=0x0000014b233e2000 nid=0xdf18 runnable

"GC Thread#0" os_prio=2 cpu=0.00ms elapsed=69.36s tid=0x0000014b7f051000 nid=0xe9d0 runnable

"G1 Main Marker" os_prio=2 cpu=0.00ms elapsed=69.36s tid=0x0000014b7f0aa800 nid=0xec8c
  runnable

"G1 Conc#0" os_prio=2 cpu=0.00ms elapsed=69.36s tid=0x0000014b7f0ac800 nid=0xb118 runnable

"G1 Refine#0" os_prio=2 cpu=0.00ms elapsed=69.36s tid=0x0000014b232f5000 nid=0xe12c runnable

"G1 Young RemSet Sampling" os_prio=2 cpu=0.00ms elapsed=69.36s tid=0x0000014b232f6000 nid=0
  x9aa4 runnable

"VM Periodic Task Thread" os_prio=2 cpu=15.62ms elapsed=69.22s tid=0x0000014b24028800 nid=0
  x8448 waiting on condition

JNI global refs: 24, weak refs: 0

PS C:\Users\dan>

```

места, по которым видно дедлок

```

"Thread-0"... java.lang.Thread.State: WAITING (on object monitor)
at java.lang.Thread.join(...)
at two_threads.lambda$main$0(two_threads.java:10)

```

```

"Thread-1"... java.lang.Thread.State: WAITING (on object monitor)
at java.lang.Thread.join(...)
at two_threads.lambda$main$1(two_threads.java:16)

```

почему это подтверждает дедлок

Thread-0 и Thread-1 одновременно находятся в состоянии WAITING и оба блокируются внутри Thread.join() (Thread.join(...)). Thread-0 (two_threads.java:10) ожидает завершения Thread-1, а Thread-1 (two_threads.java:16) ожидает завершения Thread-0. так образуется цикл ожидания в графе зависимостей:

Thread-0 → Thread-1 → Thread-0